

There is a discrepancy between the code and formula (9) in the paper. Formula (9) describes the formula for the D-matrix and consists of three parts. In the following, the formula is reconstructed from the R code and compared with formula (9) in the paper.

1 part 1 - "denom":

Denom is the first part of formula (9). In the paper this is:

$$Denom^{-1} = \sum_i (I - H_{ii})^{\otimes 2} + \sum_i \sum_{j \neq i} H_{ij}^{\otimes 2}$$

In the code this is:

$$\begin{aligned} Denom^{-1} &= I \otimes I - \sum_i w_i (I \otimes H_{ii}) - \sum_i w_i (H_{ii} \otimes I) + \sum_i (K_{mi})^{\otimes 2} \sum_i w_i (HH_i)^{\otimes 2} \\ &= \sum_i w_i (I \otimes I) - \sum_i w_i (I \otimes H_{ii}) - \sum_i w_i (H_{ii} \otimes I) + \sum_i \sum_j (K_{mi})^{\otimes 2} w_j (HH_j)^{\otimes 2} \\ &= \sum_i w_i (I \otimes I) - w_i (I \otimes H_{ii}) - w_i (H_{ii} \otimes I) + \sum_i \sum_j w_j (K_{mi} HH_j)^{\otimes 2} \\ &= \sum_i w_i (I \otimes I - I \otimes H_{ii} - H_{ii} \otimes I) + \sum_i w_i (K_{mi} HH_i)^{\otimes 2} + \sum_i \sum_{j \neq i} w_j (K_{mi} HH_j)^{\otimes 2} \\ &= \sum_i w_i (I \otimes I - I \otimes H_{ii} - H_{ii} \otimes I + (H_{ii})^{\otimes 2}) + \sum_i \sum_{j \neq i} w_j (K_{mi} HH_j)^{\otimes 2} \\ &= \sum_i w_i (I - H_{ii})^{\otimes 2} + \sum_i \sum_{j \neq i} w_j H_{ij}^{\otimes 2} \end{aligned}$$

with: $H_{ij} = K_{mi} HH_j$

2 part 2 - "vec(Sb)":

In the paper this is:

$$vec(S_b) = \sum_i b_i b'_i$$

In the code this is:

$$vec(S_b) = \sum_i w_i b_i b'_i$$

3 part 3 - "vec(c)":

In the paper this is:

$$vec(c) = \sum_i \left[(I - H_{ii})^{\otimes 2} + \sum_{j \neq i} H_{ij}^{\otimes 2} \right] vec(R_i)$$

In the code this is:

$$\begin{aligned} vec(c) &= vec\left(\sum_i R_i w_i - \sum_i H_{ii} R_i w_i - \sum_i R_i w_i H_{ii} + \sum_i K_i H H_i R_i w_i H H_i' K_{mi}'\right) \\ &= vec\left(\sum_i (R_i w_i - H_{ii} R_i w_i - R_i w_i H_{ii}) + \sum_i H_{ii} R_i w_i H_{ii}'\right) \\ &= \sum_i (vec(R_i w_i) - vec(H_{ii} R_i w_i) - vec(R_i w_i H_{ii})) + \sum_i vec(H_{ii} R_i w_i H_{ii}') \\ &= \sum_i ((I \otimes I) vec(R_i w_i) - (I \otimes H_{ii}) vec(R_i w_i) - (H_{ii}' \otimes I) vec(R_i w_i) + (H_{ii}' \otimes H_{ii}') vec(R_i w_i)) \\ &= \sum_i ((I \otimes I) - (I \otimes H_{ii}) - (H_{ii}' \otimes I) + (H_{ii}' \otimes H_{ii}')) w_i vec(R_i) \\ &= \sum_i (I - H_{ii})^{\otimes 2} w_i vec(R_i) \end{aligned}$$

with: $R_i = \tilde{\Sigma} \otimes (Z_i' Z_i)^{-1}$ and H_{ii} a diagonal matrix

There are 2 differences:

- The code includes the weighting factor w_i . In the case where $w_i = \frac{1}{N_{clusters}}$ this would not matter, but it does matter for other weighting schemes.
- For $vec(c)$ the code misses $\sum_{k \neq i} H_{ik}^{\otimes 2}$